Study guide: Paper 009

AuthorityEffect: None. The Apartment 50 preface is fictional framing. Attempt the quiz before reading the instructor key. No learner study or qualification is reported.

Route through the paper

Level	Read	Practice	Move on when you can
1	Section 1	Reduce a broad claim	Separate assumption, witness, execution, and obligation
2	Section 3 and notebook	Predict both operations	Explain the invariant and identify changed input
3	Section 4	Corrupt controls separately	Distinguish feasibility, closure, objective matching, and digest binding
4	Section 4.1 and trace	Reconstruct the four cells	Give numerators, denominators, and label provenance
5	Sections 5–6 and Lean	Read definitions before names	Distinguish formal facts, census scope, and runtime correspondence
6	Appendix A.1–A.4	Follow each verdict to its operation	Identify four families and five probes
7	Appendix A.5	Inspect corrections	Separate necessary evidence from guarantees about a checker
8	Sections 6–8	Write a scoped record	Preserve refusals, missing bridges, and unexecuted dependencies

Prepare the exact calculation

Use edges (0,1), (1,2), (2,0). Write all three endpoint differences before subtracting. Each potential occurs once with each sign around the cycle; the differences sum to zero. This cancellation explains every potential on the declared cycle. A graph path is not an automatically filled simplex.

Replacing claims by df changes the input. Subtracting df from fixed claims preserves their cycle sum. Save original and transformed inputs: a low score on replacement data does not show the original obstruction disappeared.

Predict the displayed numbers, then run:

```
python tools/replay_notebook.py
```

The default executes all five ordinary Python cells in a fresh CPython process, checks assertions, and compares stored outputs without rewriting the notebook. Jupyter kernel execution is a separate `--engine jupyter` check. Neither mode runs the native graph checker or grades a learner's explanation.

Prepare the historical recount

Read evidence provenance. "Refused" and "not refused" mean the original checker's decisions, not physical ground truth. Separate CHALLENGE and CONCUR, check all cells total ten, and keep a later checker decision as a distinct observation.

False-positive fraction divides challenges of not-refused proposals by all not-refused proposals. Precision divides challenged refused proposals by all challenges. Different denominators answer different questions. Use exact fractions for the new recount; zero denominators are undefined.

```
python tools/review_capstone.py
python -m pytest -q
```

The capstone verifies retained digests and the complete condition census, changes one verdict in a copy, and ignores the constant caught flag. Tests also reject non-Boolean labels, unknown verdicts, and label/status disagreement. The gate record inspection does not execute the gate. Complete the written report too.

Diagnose a missed answer

Mistake	Return to	Correction
A hash or token is treated as proof	Level 1	State the missing property check
Subtraction is confused with replacement	Level 2	Track original input, operation, residual, and sum

Mistake	Return to	Correction
Equal LP objectives replace feasibility	Level 3	Check both feasible regions
The caught flag yields perfect detection	Level 4	Count actual decisions against historical labels
A theorem name certifies a runtime	Level 5	Identify the correspondence obligation
Four families are called four records	Level 6	Count F2a and F2b separately
A bound receipt makes a critique true	Level 7	Separate identity, authentication, and validity
A retained report is called a fresh replay	Level 8	Record the command actually run

Write a review record

Include claim text, source revision, artifact path/digest, assumptions, exact input, command, environment or executable pin, observed result, review disposition, and remaining obligation. Mark a command `NOT_RUN` until observed. The manuscript's dispositions are review terms, not a new runtime API.

The v0.1 review record separates fresh execution from earlier reports. Deposit and ZIP checks establish file custody, not the truth of every answer. A pinned Lean build is a separate dependency check.